

Molecular Biology & Genetics

STUDY NOTES

BOTANY & BIOLOGY

This comprehensive study guide covers the core principles of molecular biology and genetics — from the molecular machinery of DNA replication, transcription, and translation to the genetic code, gene regulation, and modern biotechnology tools including PCR, blotting techniques, and CRISPR. These notes are designed for undergraduate and graduate biology students preparing for exams, with precise technical language, visual diagrams, and exam-focused summaries for each topic.

DNA Replication

MOLECULAR BIOLOGY

DNA replication is the semi-conservative process by which a cell duplicates its genetic material before cell division. Each daughter molecule contains one original (parental) strand and one newly synthesized strand. Replication occurs during the **S-phase** of the cell cycle and proceeds in the **5' → 3' direction**. The process requires a template strand, dNTPs (deoxyribonucleotide triphosphates), and a suite of enzymes working in concert.

Key Enzymes & Proteins

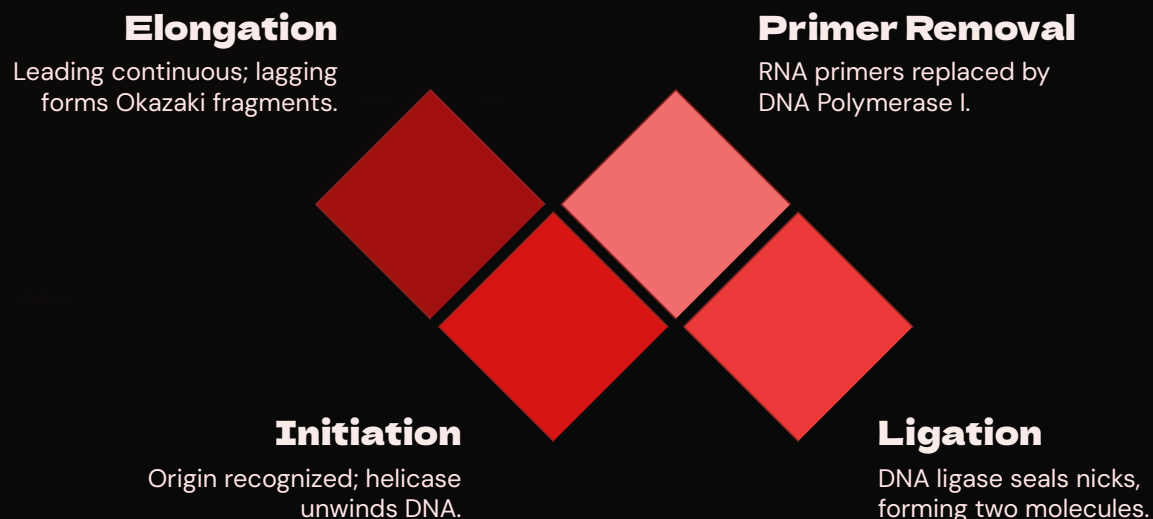
- **Helicase** — unwinds the double helix at the replication fork
- **SSB Proteins** — stabilize single-stranded DNA
- **Topoisomerase** — relieves supercoiling ahead of the fork
- **Primase** — synthesizes RNA primers (~10 nucleotides)
- **DNA Polymerase III** — main elongation enzyme (prokaryotes)
- **DNA Polymerase I** — removes RNA primers, fills gaps
- **DNA Ligase** — joins Okazaki fragments on lagging strand
- **Telomerase** — extends telomeres in eukaryotes

Leading vs. Lagging Strand

- **Leading strand:** synthesized continuously toward the replication fork
- **Lagging strand:** synthesized discontinuously as Okazaki fragments (1000–2000 nt in prokaryotes; 100–200 nt in eukaryotes)
- Both strands are synthesized 5'→3'; the lagging strand loops to allow polymerase to work in the same physical direction

Replication Fidelity

- 3'→5' exonuclease proofreading activity
- Error rate: ~1 in 10⁹ bases after proofreading & mismatch repair
- Mismatch Repair (MMR) corrects post-replication errors



Replication initiates at specific **origins of replication (ori)**. Prokaryotes have a single ori; eukaryotes have thousands. The **replisome** is the multi-protein complex that carries out replication at each fork. In eukaryotes, replication timing is regulated — euchromatin replicates early, heterochromatin replicates late.

Transcription & Translation

CENTRAL DOGMA

The **Central Dogma** of molecular biology describes the flow of genetic information: **DNA → RNA → Protein**. Transcription converts DNA into mRNA; translation decodes mRNA into a polypeptide chain. In prokaryotes, both processes occur simultaneously in the cytoplasm (coupled transcription-translation). In eukaryotes, transcription occurs in the nucleus and translation in the cytoplasm, with mRNA processing as an intermediate step.

Transcription

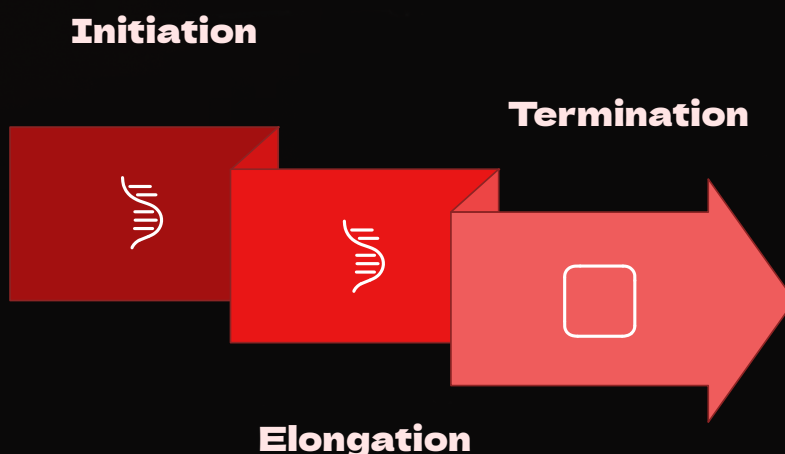
Synthesis of RNA from a DNA template by **RNA Polymerase**.

- **Promoter:** TATA box (eukaryotes), Pribnow box -10 (prokaryotes)
- **Sigma factor (σ):** directs RNA pol to promoter in bacteria
- **Elongation:** RNA synthesized 5'→3'; DNA rewinds behind
- **Termination:** Rho-dependent or Rho-independent (hairpin)
- **Eukaryotic processing:** 5' capping, 3' poly-A tail, splicing (spliceosome removes introns)
- **Types of RNA:** mRNA, tRNA, rRNA, snRNA, miRNA

Translation

Decoding of mRNA into protein at the **ribosome**.

- **Ribosome:** 70S (prokaryotes: 50S+30S), 80S (eukaryotes: 60S+40S)
- **tRNA:** carries amino acids; anticodon pairs with mRNA codon
- **Aminoacyl-tRNA synthetase:** charges tRNA with correct amino acid
- **Sites:** A (aminoacyl), P (peptidyl), E (exit)
- **Initiation:** AUG start codon; fMet-tRNA in prokaryotes, Met-tRNA in eukaryotes
- **Elongation:** peptide bond formation by peptidyl transferase (rRNA ribozyme)
- **Termination:** UAA, UAG, UGA stop codons; release factors (RF)



Translation proceeds through three stages — initiation, elongation, and termination. The ribosome reads mRNA codons in triplets, matching each with the appropriate aminoacyl-tRNA. Post-translational modifications (phosphorylation, glycosylation, folding) further process the polypeptide into a functional protein.

Operon Concept & Gene Regulation

GENE REGULATION

An **operon** is a functional unit of DNA in prokaryotes containing a cluster of genes under the control of a single promoter. The operon model was proposed by **Jacob and Monod (1961)** based on studies of *E. coli*. Operons allow coordinated expression of functionally related genes. Eukaryotes do not have operons but use enhancers, silencers, and transcription factors for regulation.

Lac Operon (Inducible)

- **Structural genes:** lacZ (β -galactosidase), lacY (permease), lacA (transacetylase)
- **Repressor (lacI):** binds operator in absence of lactose $\rightarrow$ OFF
- **Inducer:** allolactose binds repressor $\rightarrow$ releases operator $\rightarrow$ ON
- **CAP-cAMP:** positive regulation when glucose is low
- **Dual control:** negative (repressor) + positive (CAP)

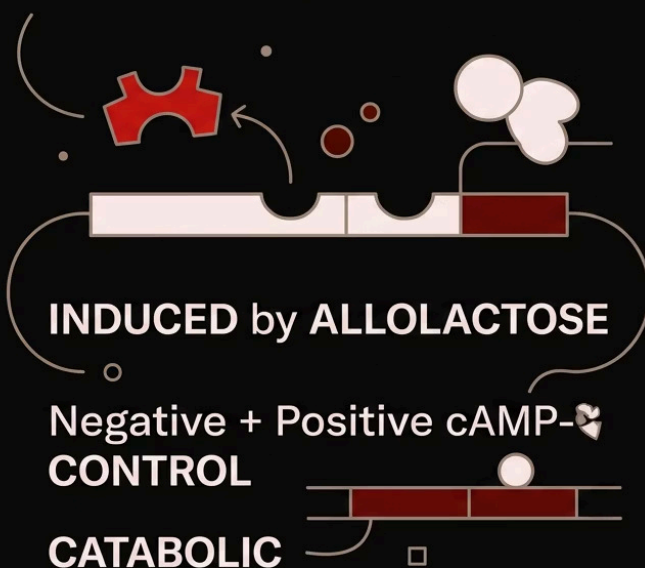
Trp Operon (Repressible)

- **Structural genes:** trpE, trpD, trpC, trpB, trpA (tryptophan biosynthesis)
- **Repressor:** inactive alone; active when bound to tryptophan (corepressor)
- **High Trp:** repressor-corepressor complex binds operator $\rightarrow$ OFF
- **Attenuation:** premature termination via leader peptide & hairpin formation
- **Feedback:** end-product regulates its own synthesis

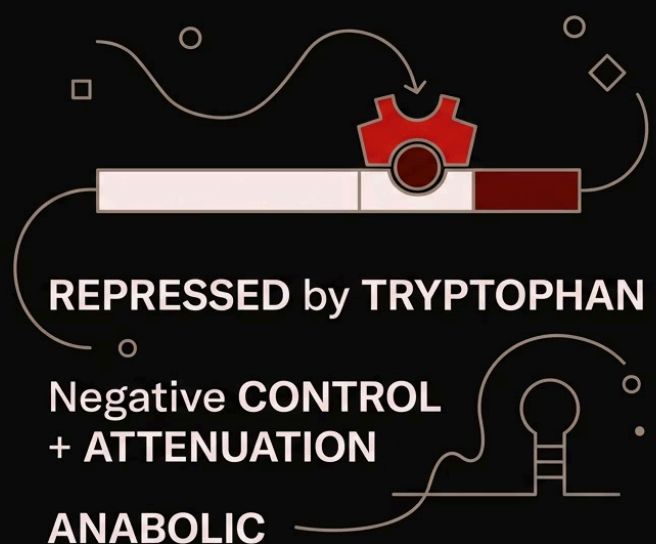
Eukaryotic Regulation

- **Chromatin remodeling:** histone acetylation (open), methylation (variable)
- **Transcription factors:** activators, repressors, co-activators
- **Enhancers/Silencers:** distal regulatory elements
- **RNA interference:** miRNA and siRNA post-transcriptional silencing
- **Epigenetics:** DNA methylation, histone modification, non-coding RNA

LAC OPERON INDUCIBLE (Normally OFF)



TRP OPERON REPRESSIBLE (Normally ON)



Genetic Code & Mutations

GENETICS

The **genetic code** is the set of rules by which information encoded in mRNA is translated into proteins. It is nearly universal across all organisms, with minor variations in mitochondria and some protists.

Understanding the code and the types of mutations that alter it is fundamental to molecular genetics and evolutionary biology.

Properties of the Genetic Code

- **Triplet:** 3 nucleotides = 1 codon (64 codons total)
- **Degenerate/Redundant:** multiple codons encode the same amino acid (e.g., Leu has 6 codons)
- **Unambiguous:** each codon specifies only one amino acid
- **Non-overlapping:** codons are read sequentially without overlap
- **Commaless:** no gaps between codons
- **Universal:** same in almost all organisms
- **Start codon:** AUG (Met/fMet)
- **Stop codons:** UAA (ochre), UAG (amber), UGA (opal)
- **Wobble hypothesis:** third base pairing is flexible (Crick, 1966)

Types of Mutations

- **Point mutations:** single base change
 - — **Silent:** no amino acid change (degeneracy)
 - — **Missense:** different amino acid (e.g., sickle cell: Glu→Val)
 - — **Nonsense:** creates premature stop codon
- **Frameshift:** insertion or deletion shifts reading frame
- **Transition:** purine ↔ purine or pyrimidine ↔ pyrimidine
- **Transversion:** purine ↔ pyrimidine
- **Causes:** UV (pyrimidine dimers), chemicals, radiation, replication errors
- **Repair:** photoreactivation, nucleotide excision repair (NER), base excision repair (BER)

64

Total Codons

61 sense + 3 stop codons

20

Amino Acids

Encoded by the standard genetic code

3

Stop Codons

UAA, UAG, UGA terminate translation

1

Start Codon

AUG codes for Methionine

Mendelian Genetics & Linkage

CLASSICAL GENETICS

Gregor Mendel (1865) established the fundamental laws of inheritance through pea plant experiments. His work, rediscovered in 1900, forms the foundation of classical genetics. Mendel's laws describe how traits are transmitted from parents to offspring through discrete hereditary units (genes).

1

Law of Dominance

In a heterozygote, one allele (dominant) masks the expression of the other (recessive).
Dominant allele expressed in both homozygous and heterozygous conditions.

2

Law of Segregation

Allele pairs separate during gamete formation (meiosis I). Each gamete receives only one allele.
Explains 3:1 phenotypic ratio in F_2 monohybrid cross.

3

Law of Independent Assortment

Alleles of different genes assort independently during gamete formation (applies to genes on different chromosomes). Produces 9:3:3:1 ratio in F_2 dihybrid cross.

4

Exceptions & Extensions

Incomplete dominance, codominance, multiple alleles (ABO blood group), epistasis, pleiotropy, polygenic inheritance, and sex-linked traits modify Mendelian ratios.

Linkage & Crossing Over

- **Linkage:** genes on the same chromosome tend to be inherited together
- **Complete linkage:** no crossing over; only parental types
- **Incomplete linkage:** crossing over produces recombinants
- **Crossing over:** exchange of segments between homologous chromosomes at chiasmata (prophase I)
- **Recombination frequency (RF):** % recombinants = map distance
- **1% RF = 1 map unit (m.u.) = 1 centiMorgan (cM)**
- **Interference (I):** $1 - \text{coefficient of coincidence}$; reduces double crossovers

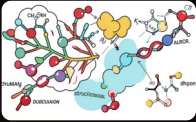
Gene Mapping

- **Two-point cross:** distance between two genes
- **Three-point cross:** determines gene order and distances; most accurate
- **Parental types:** most frequent; **Double crossovers:** least frequent
- **Mapping function:** Haldane or Kosambi (corrects for multiple crossovers)
- Linked genes violate independent assortment — $RF < 50\%$
- $RF = 50\%$ indicates genes are either on different chromosomes or very far apart

Chromosomal Aberrations

CYTOGENETICS

Chromosomal aberrations are structural or numerical changes in chromosomes that can cause genetic disorders, developmental abnormalities, or evolutionary change. They arise from errors in meiosis, mitosis, or exposure to mutagens (radiation, chemicals). Karyotyping is the primary diagnostic tool.



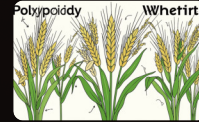
Structural Aberrations

- **Deletion:** loss of chromosome segment (e.g., Cri-du-chat: 5p-)
- **Duplication:** extra copy of segment; gene dosage effect
- **Inversion:** segment reversed (paracentric or pericentric)
- **Translocation:** segment moves to non-homologous chromosome
 - — **Reciprocal:** mutual exchange
 - — **Robertsonian:** fusion of acrocentric chromosomes



Numerical Aberrations

- **Aneuploidy:** gain or loss of individual chromosomes
 - — **Nullisomy:** $2n-2$ (both homologs lost)
 - — **Monosomy:** $2n-1$ (e.g., Turner syndrome: 45,X)
 - — **Trisomy:** $2n+1$ (e.g., Down syndrome: 21; Klinefelter: XXY)
- **Euploidy:** change in complete chromosome sets
 - — **Haploid (n), Diploid (2n), Polyploid (3n, 4n...)**
- **Polyploidy** common in plants; rare lethal in animals



Significance in Plants

- **Autopolyploidy:** genome duplication within species (e.g., seedless watermelon $3n$)
- **Allopolyploidy:** hybridization + duplication between species (e.g., wheat: hexaploid $6n$)
- **Advantages:** larger organs, hybrid vigor, seedlessness
- **Colchicine:** induces polyploidy by inhibiting spindle formation
- ~70% of angiosperms have polyploid ancestry

⚠ Clinical Note: Non-disjunction during meiosis I or II is the primary cause of aneuploidy. Advanced maternal age increases risk of trisomy 21 (Down syndrome). Prenatal diagnosis via amniocentesis or chorionic villus sampling (CVS) detects chromosomal abnormalities.

Population Genetics

EVOLUTIONARY BIOLOGY

Population genetics studies the distribution and changes in allele frequencies within populations under the influence of evolutionary forces. It bridges Mendelian genetics with Darwinian evolution, providing a mathematical framework for understanding how populations evolve over time.



Hardy-Weinberg Equilibrium

In a large, randomly mating population with no evolutionary forces, allele and genotype frequencies remain constant.

- $p^2 + 2pq + q^2 = 1$ (genotype frequencies)
- $p + q = 1$ (allele frequencies)
- Assumptions: no mutation, no migration, no selection, infinite population, random mating
- Deviation from HWE indicates evolutionary change



Evolutionary Forces

- **Mutation:** introduces new alleles; ultimate source of variation
- **Natural Selection:** differential survival/reproduction; directional, stabilizing, or disruptive
- **Gene Flow (Migration):** allele exchange between populations; homogenizes frequencies
- **Genetic Drift:** random allele frequency changes; stronger in small populations
 - **Bottleneck effect:** drastic population reduction
 - **Founder effect:** new population from few individuals



Quantitative Measures

- **Allele frequency:** $p = (2N_{AA} + N_{Aa}) / 2N$
- **Heterozygosity (H):** expected = $2pq$; observed vs. expected indicates inbreeding
- **F-statistics (F_{ST}):** measures population differentiation
- **Fitness (w):** relative reproductive success; $w = 1 - s$ (selection coefficient)
- **Heritability (h^2):** proportion of phenotypic variance due to genetics

100%

HWE Sum

$p^2 + 2pq + q^2 = 1$ at equilibrium

50%

Max Heterozygosity

When $p = q = 0.5$, $2pq$ is maximized

1%

Typical Mutation Rate

$\sim 10^{-8}$ to 10^{-9} per base per generation

Recombinant DNA Technology & PCR

BIOTECHNOLOGY

Recombinant DNA (rDNA) technology involves cutting, joining, and cloning DNA from different sources to create novel genetic combinations. Developed in the 1970s (Cohen & Boyer, 1973), it revolutionized medicine, agriculture, and research. **Polymerase Chain Reaction (PCR)**, invented by Kary Mullis (1983), amplifies specific DNA sequences exponentially without cloning.

rDNA Technology Steps

- **Isolation:** extract DNA from source organism
- **Cutting:** restriction endonucleases (e.g., EcoRI, HindIII) create sticky or blunt ends
- **Ligation:** DNA ligase joins insert + vector (plasmid, phage, cosmid, BAC, YAC)
- **Transformation:** recombinant DNA introduced into host (E. coli, yeast)
- **Selection:** antibiotic resistance markers, blue-white screening (lacZ)
- **Expression:** host produces recombinant protein (e.g., insulin, Bt toxin)

Key Tools

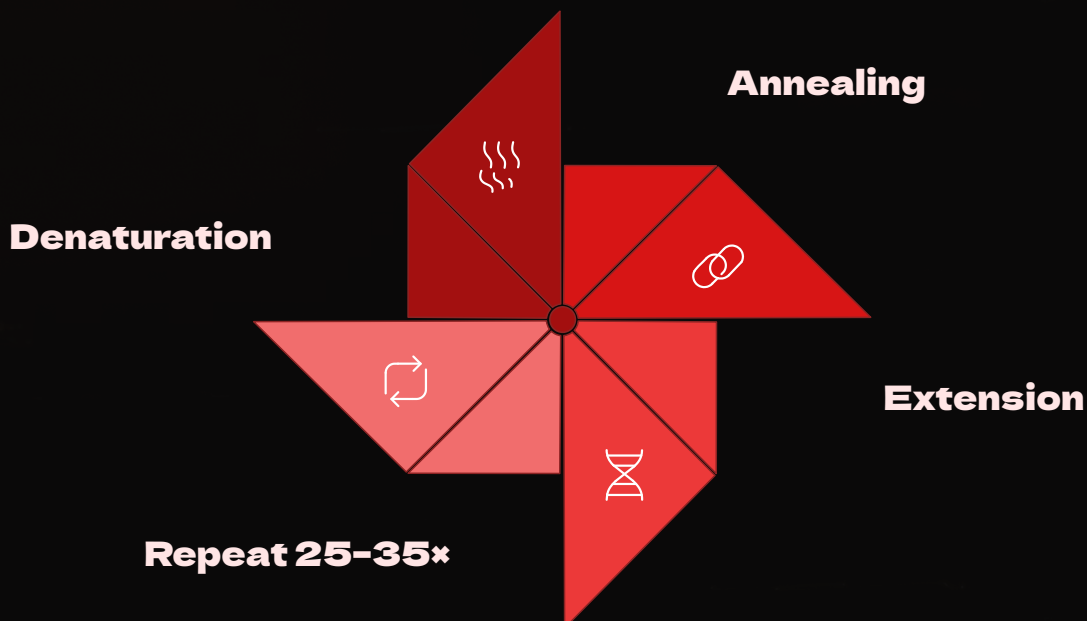
- **Vectors:** pUC19, pBR322 (plasmids); λ phage; cosmids
- **Hosts:** E. coli, Agrobacterium (plants), yeast
- **Libraries:** genomic vs. cDNA libraries

PCR – Polymerase Chain Reaction

- **Components:** template DNA, primers (forward + reverse), Taq polymerase, dNTPs, buffer
- **Denaturation:** 94–95°C — strands separate
- **Annealing:** 50–65°C — primers bind to target
- **Extension:** 72°C — Taq synthesizes new strand
- **Amplification:** 2^n copies after n cycles (30 cycles = $\sim 10^9$ copies)
- **Variants:** RT-PCR (RNA template), qPCR (quantitative), multiplex PCR
- **Applications:** forensics, diagnostics, phylogenetics, cloning

blotting Techniques

- **Southern blot:** DNA detection (restriction fragments + probe)
- **Northern blot:** RNA/mRNA detection (gene expression)
- **Western blot:** protein detection (antibodies)
- **Eastern blot:** post-translational modifications



Each PCR cycle doubles the target DNA, producing exponential amplification. After 30 cycles, a single template molecule can generate over a billion copies. Taq polymerase, isolated from *Thermus aquaticus*, is thermostable and survives repeated heating cycles.

CRISPR-Cas9 & Modern Gene Editing

ADVANCED BIOTECHNOLOGY

CRISPR-Cas9 (Clustered Regularly Interspaced Short Palindromic Repeats) is a revolutionary gene-editing technology adapted from a bacterial immune system. Discovered by Jennifer Doudna and Emmanuelle Charpentier (Nobel Prize, 2020), it enables precise, programmable DNA modifications in virtually any organism. CRISPR has transformed research, medicine, and agriculture.

1 Mechanism of CRISPR-Cas9

The system uses a **guide RNA (gRNA)** to direct the **Cas9 endonuclease** to a specific DNA sequence. Cas9 creates a double-strand break (DSB) at the target site. The cell repairs the break via:

- **NHEJ (Non-Homologous End Joining):** error-prone; creates insertions/deletions (indels) → gene knockout
- **HDR (Homology-Directed Repair):** uses donor template → precise gene insertion or correction
- **PAM sequence:** NGG (for SpCas9) required adjacent to target for Cas9 binding

2 Applications

- **Medicine:** sickle cell disease, β -thalassemia, cancer immunotherapy (CAR-T), HIV
- **Agriculture:** disease-resistant crops, improved yield, reduced allergens
- **Research:** gene knockout/knockin, epigenetic editing (dCas9 fusions)
- **Gene drive:** spreads edited genes through populations (malaria mosquito control)

3 Advanced Variants & Ethics

- **Base editing:** single-base changes without DSB (C→T, A→G)
- **Prime editing:** "search-and-replace" without DSB or donor template
- **CRISPRi/CRISPRa:** dCas9 represses or activates gene expression
- **Ethical concerns:** off-target effects, germline editing, designer babies, ecological impact of gene drives
- **Regulation:** varies globally; WHO and NAS provide guidelines

Southern Blot

Detects specific DNA sequences. DNA digested, separated by gel electrophoresis, transferred to membrane, hybridized with labeled probe.

Northern Blot

Detects specific RNA/mRNA. Measures gene expression levels and transcript size. RNA separated, transferred, probed.

Western Blot

Detects specific proteins. Proteins separated by SDS-PAGE, transferred, detected with primary + secondary antibodies.

CRISPR Summary

gRNA guides Cas9 to target → DSB → NHEJ (knockout) or HDR (knockin). PAM sequence (NGG) required. Base/prime editing reduce off-target effects.

Exam Tip: Remember the blotting mnemonic — Southern = DNA, Northern = RNA, Western = Protein. CRISPR questions often test the difference between NHEJ (indels, knockout) and HDR (precise edit, requires template). Always mention PAM sequence requirement for Cas9 activity.